群測感知技術與應用 (Crowd sensing)
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Mobile Crowd Sensing : Traffic Monitoring
Social Interaction
Epidemic Disease Monitoring
Things Idenification
Environment Monitoring
Reporting From Disater Sitaations
Special Applications
Urban Dynamic Mining
logistics Tracking

* Modern user-centric devices can sense, collect and analyze data

* Mobile crowdsensing (MSC) aims to exploit data from the aggregate contribution of mobile users

* MSC [ A service provider
  Mobile users

* Incentive : mobile users provide data and get reward

* Three forms of commonly used rewards : (1) Service experience
  (2) Reputation credit
  (3) Monetary token

* From the economic & solo-economic perspective

* Game theory → game-based incentives
* factor: quality, reputation, mallcious, requirment, maxium reward

* Incentive-G Design:
  1. Accomplishing crowdsensing tasks
  2. Analyzing sensing data
  3. Updating reputation values
  4. Updating rewards for users

  - Data analysis process
  - Reputation-intiated process
  - Voting-based decision process

* Data quality supervision
  - data sreening
  - data similary

* Revenue Formulation in incentive-G Design

* Total revenue and reward w.r.t. a service provider

* Urban sensing data: obtain multi-source urban sensing data in smart cities.

AI/ML/Big data
- Intelligent Transporatation System
- Internet of Things
- Mobile Crowdsensing